ESP32-based Smart Purifier with IoT Integration

ASE Project1 Guide **(Vishwesh Deshmukh)**,Pratibha Chavan,Purva Chavan,Srushti Chavan,Shantanu Chavre,Ridham Chhajer,Manthan Chidrewar

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ABSTRACT

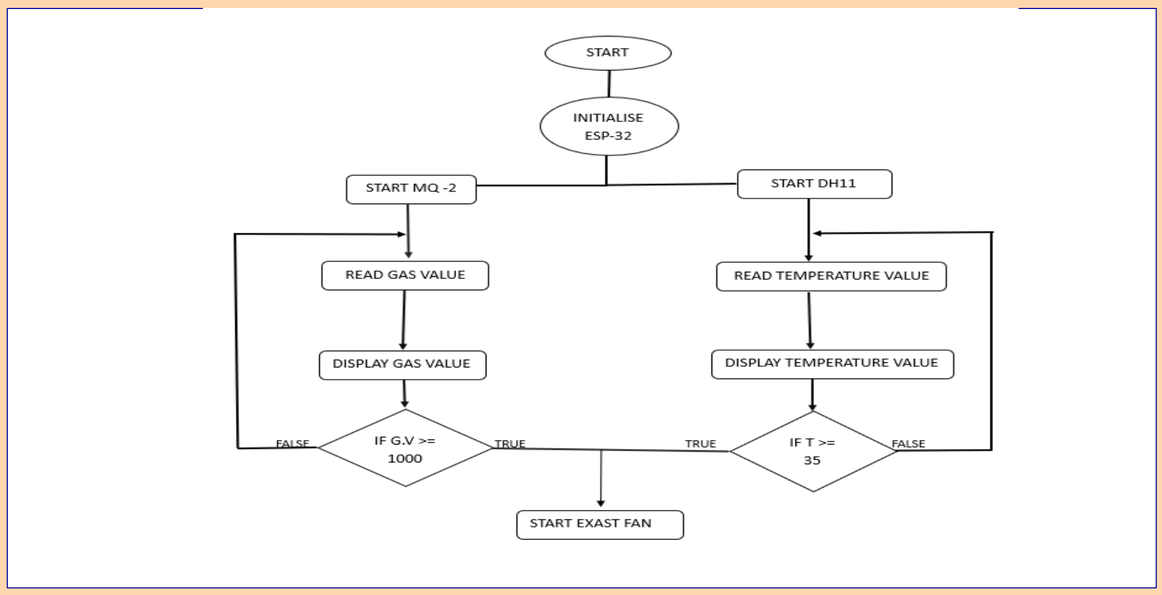
- ✓ Air Pollution Crisis : A serious threat,especially for vulnerable groups.
- ✓ Health Risks : Leads to thyroid issues,lung cancer,respiratory diseases, and reduces lifespan.
- ✓ Smart Air Purifier : Monitor AQI and auto-adjusts for better air quality.
- ✓ Sensors: ●MQ-2 : Detects smoke,CO_2,NO_2.
  ●MQ-3 : Detects CO,Methane,Benzene.
- ✓ ESP32 Microcontroller : Collects real-time data and updates a filter control website.
- ✓ Impact : Enhances indoor air quality and saves energy efficiently.

INTRODUCTION

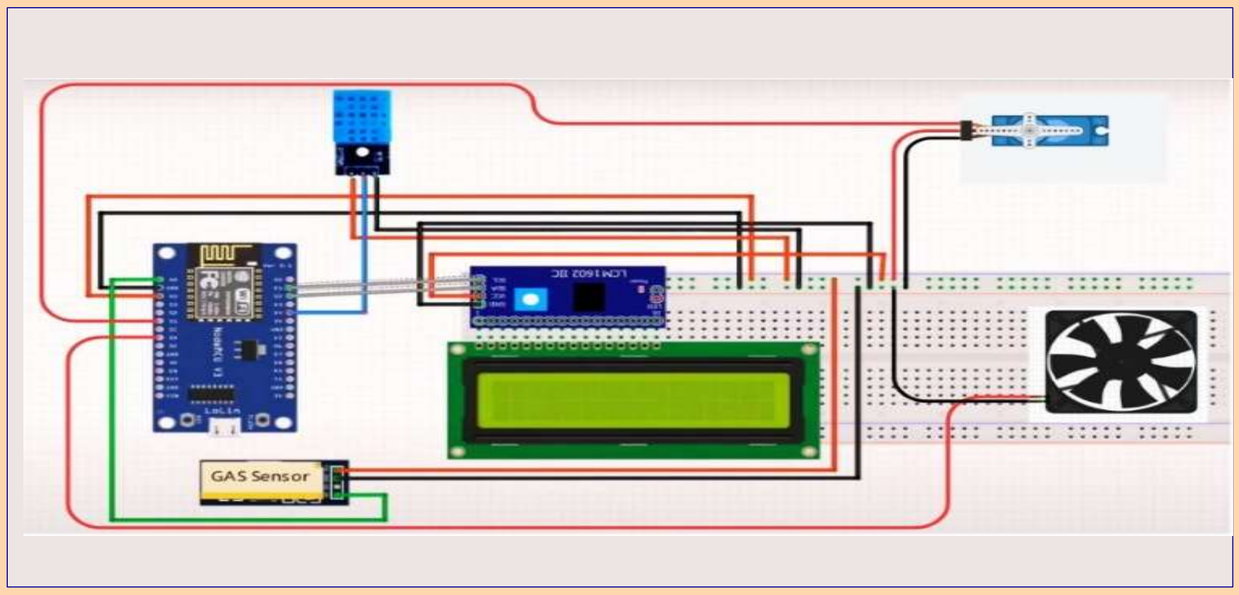
- ✓ Air Pollution Concern:A serious issue affecting indoor and outdoor air quality
- ✓ Efficient Filteration : HEPA & Carbon Filter works together to remove dust,harmful gases and odors
  Exhaust fan ensures better air flow for cleaner air
- ✓ Smart Monitoring : MA-2DH11 Sensor constantly checks air quality and Mobile App & IOT provide realtime data & adjust purification for efficiency

METHODOLOGY

FLOW CHART



ACTUAL CIRCUIT/ DESIGNED SYSTEM



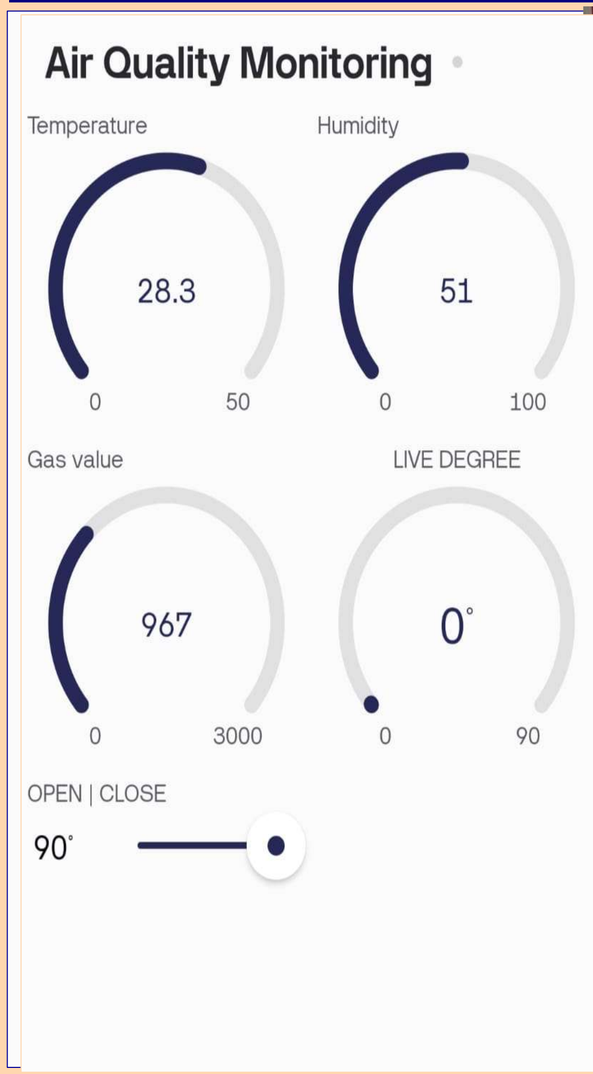
TESTING / IMPLEMENTATION

- ✓ MQ-2 and MQ-3 sensor tested for detecting CO,NO_2 and Methane.
- ✓ HEPA and Carbon filters for AQI improvement.
- ✓ ESP32 checked for real time data transmission and updates.
- ✓ Auto ON/OFF and exhaust fan tested under different AQI levels.
- ✓ Developed Prototype by assembling components and programed for AQI based control
- ✓ User monitors AQI and receive alerts via mobile app

RESULTS AND DISCUSSIONS

- ✓ Air Quality Improvement : Sensors like MQ-2 and MQ-3 can detect CO2,NO2,Methane and more pollutants and with use of HEPA and Carbon Filter particulate matter and harmful gases are removed.
- ✓ Realtime Monitoring and automation : the ESP32 microcontroller transmit air quality data to app and auto ON/OFF improves purification efficiency and optimizes energy use.
- ✓ Performance Evaluation: More responsive than traditional purifiers,logs AQI trends for better air quality management.
- ✓ Scalability : Suitable for homes,offices,hospitals and industries, also IOT integration makes it cost effective and efficient

IMP 2/3 IMAGES OF PROJECT



NOVELTY FEATURES / FINDINGS

Features
- ✓ Integration of ESP32 with AQI monitoring and Auto ON/OFF based on pollution level.
- ✓ HEPA and Carbon filter for enchanced air purification
- ✓ Real-Time AQI Data Transmission for better analysis and tracking
- ✓ Exhaust Fan that adjust its airflow dynamically for optimized purification.

Findings
- ✓ Accurate detection of harmful gases with sensors.
- ✓ Efficiently filters fine particles & gases,enhancing indoor air condition.
- ✓ Energy efficient with automation.
- ✓ Mobile app provides real-time inights that helps to track air quality changes

CONCLUSIONS

- ✓ Increasing Air Pollution severely impacts health and reduces life expectancy by increasing respiratory diseases.
- ✓ With help of Sensors and ESP32 microcontroller,harmul gases are detected and purification starts based on AQI levels.
- ✓ Efficient filteration and realtime Air Quality management enhances Indoor Air Quality.